Acute and chronic Gastritis

❖ Gastritis

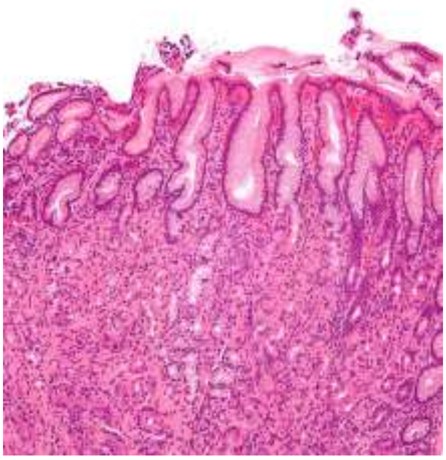
- Gastritis is a general term for a group of conditions with one thing in common: Inflammation of the lining of the stomach.
- Gastritis may occur suddenly (acute gastritis) or appear slowly over time (chronic gastritis). In some cases, gastritis can lead to ulcers and an increased risk of stomach cancer.

➢ Pathophysiology

The mechanisms of mucosal injury in gastritis involves an imbalance of **aggressive factors**, such as acid production or pepsin, and **defensive factors**, such as mucus production, bicarbonate, and blood flow.

➢ The defensive forces

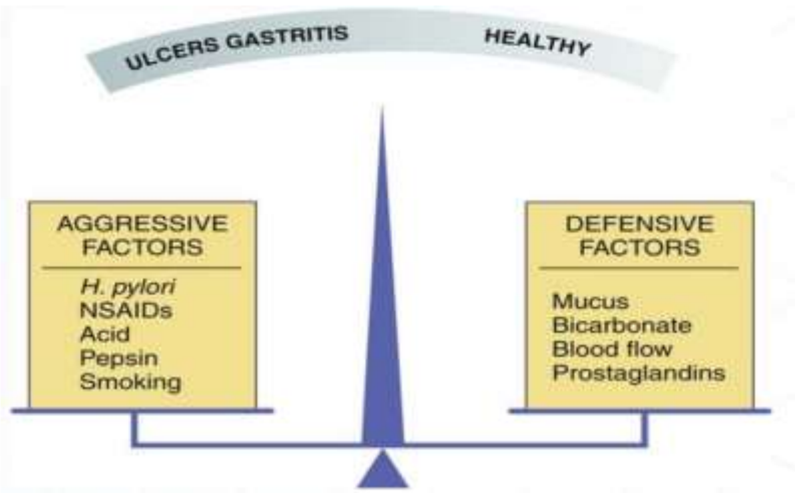
1.Bicarbonate
2.Mucus layer
3. Mucosal blood flow
4.Prostaglandins
5.Growth factors



➢ The aggressive forces

1.Helicobacter pylori
2.HCl acid
3.Pepsins
4.NSAIDs
5.Bile acids
6.Ischemia and hypoxia.
7.Smoking and alcohol.

When the aggressive factors increase or the defensive factors decrease, mucosal damage will result, leading to erosions and ulcerations.



Acute gastritis

- Several different etiologies share the same general clinical presentation; however, they differ in their unique histologic characteristics. The inflammation may involve the entire stomach (e.g., pangastritis) or a region of the stomach (e.g., antral gastritis).
- Acute gastritis can be divided into two categories: erosive (e.g., superficial erosions, deep erosions, hemorrhagic erosions) and nonerosive (generally caused by Helicobacter pylori)

Etiology

- **Drugs**: mostly Nonsteroidal anti-inflammatory drugs (NSAIDs) (eg, aspirin, ibuprofen, and naproxen). They inhibit cyclo-oxygenase enzyme and prostaglandins formation. The gastritis may occur even with small doses e. g. low dose aspirin (80 – 100mg/day) used for secondary prevention by patients with IHD. The symptoms may occur early (very shortly) after ingestion of such drugs and even after a single dose, but may be delayed for considerable time after ingestion.
  Other drugs like: cocaine; iron; colchicine
- Early stage of **H. pylori infection**: H. pylori usually cause chronic gastritis which is more important. The inflammation is usually involves the antral part of the stomach. (antral gastritis)
- **Viral infections** (eg, cytomegalovirus)
- **Fungal infections**: Candidiasis, histoplasmosis
- **Acute stress** (shock), Burn
- **Radiation**
- **Allergy and food poisoning**
- **Bile**: The reflux of bile (an alkaline medium is important for the activation of digestive enzymes in the small intestine) from the small intestine to the stomach can induce gastritis.
- **Ischemia**: This term is used to refer to damage induced by a decreased blood supply to the stomach. This rare etiology is due to the rich blood supply to the stomach.

Clinical features

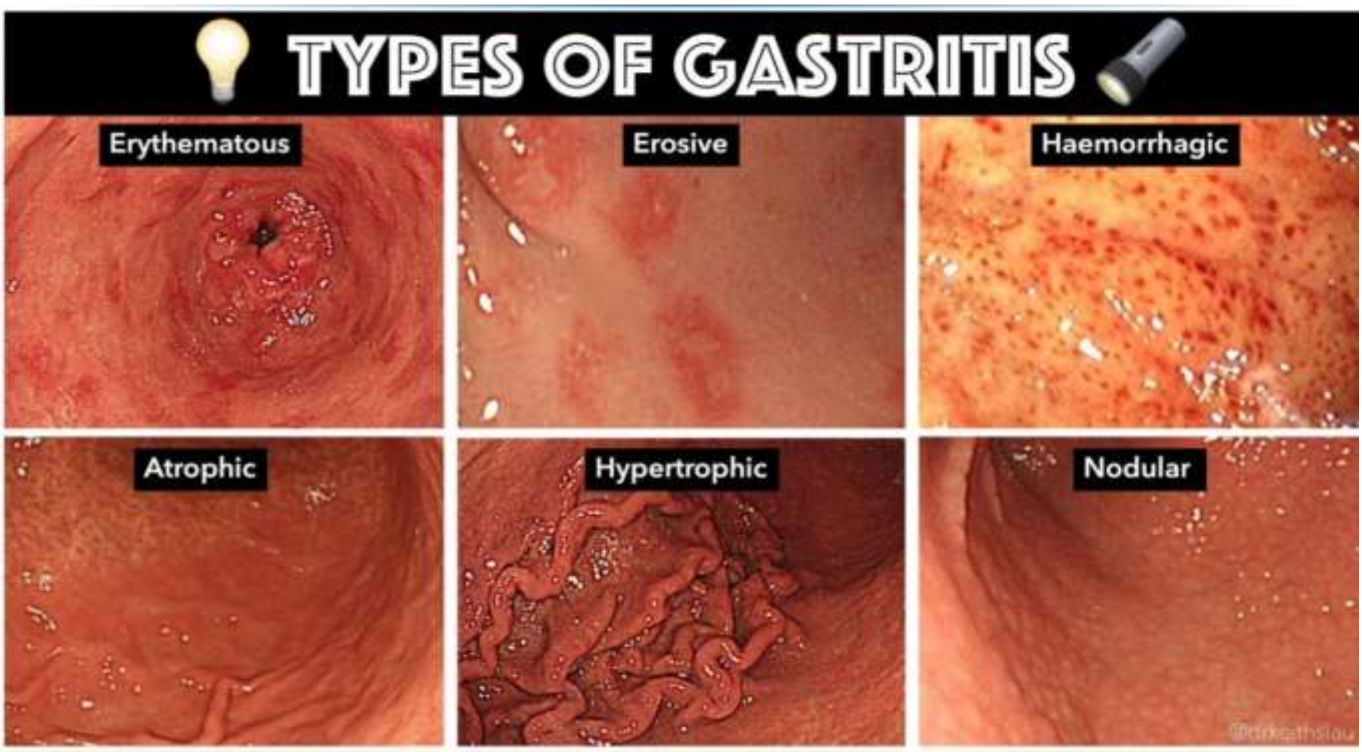
1-burning **epigastric pain** that may improve or worsen by eating.
2-nausea and/or vomiting.
3-Rarely, upper GIT bleeding.
4-The physical examination findings are often normal, with occasional mild epigastric tenderness. The examination tends to exhibit more abnormalities as the patient develops complications in relation to gastritis.

Differential Diagnosis

- Peptic ulcer disease.
- Gastric cancer.
- Lymphoma.
- Functional dyspepsia

Investigations

Diagnosis is mostly **clinical** if there is doubt or presentation prolonged **endoscopy** can be done. There is 4 endoscopic features of acute gastritis:
1-Erythematous and measles like appearance acute gastritis.
2-Erosive gastritis: characterized by presence of multiple areas of erosions. This may appear as multiple red, edematous elevated areas of mucosa with central superficial ulcers (polypoid). Usually follows NSAIDs ingestion and usually the patient is symptomatic.
3-Hemorrhagic gastritis: This is the most severe and dangerous type of gastritis which may lead to sever upper GIT bleeding. The stomach appears very red and congested and usually with multiple areas of bleeding (oozing). The mucosa is very friable and bleeds on touch.
4-Biliary gastritis.



Treatment

1.treat underlying cause.

2. Antacids and H2 blockers like ranitidine or PPI (proton-pump inhibitor) like omeprazole in usual doses for short period of time.

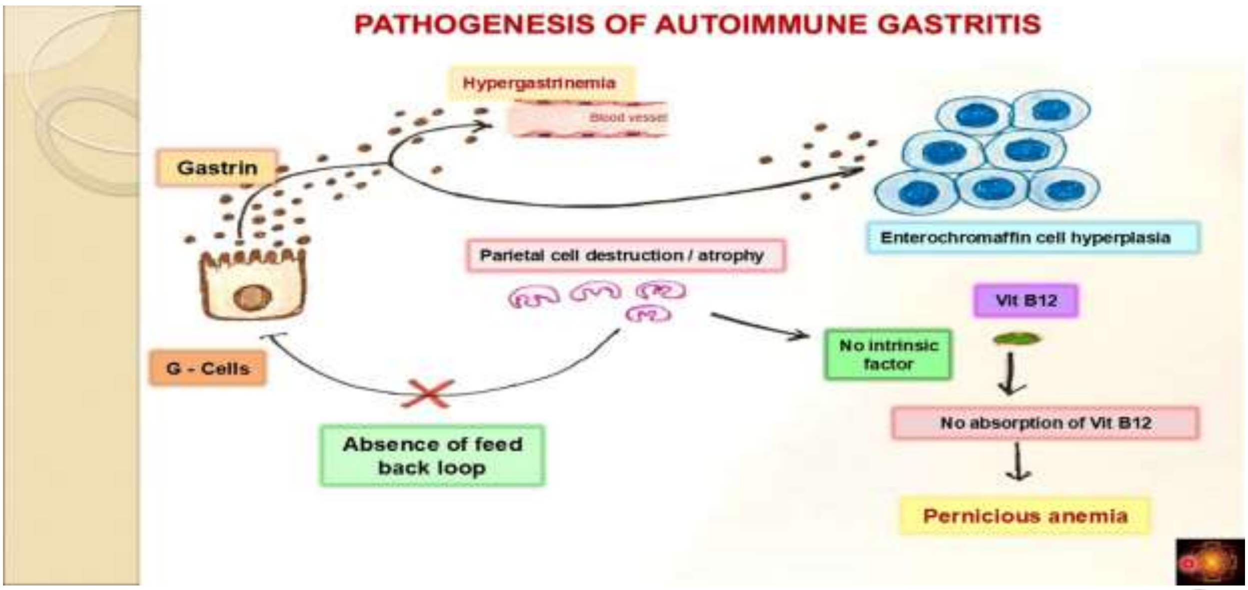
3. Anti-emetics like metoclopramide also may be needed

4. surgical interference not needed in acute gastritis except in cases of acute necrotizing gastritis, total gastrectomy may be needed to stop severe, continuous upper GIT bleeding despite other non-surgical measures.

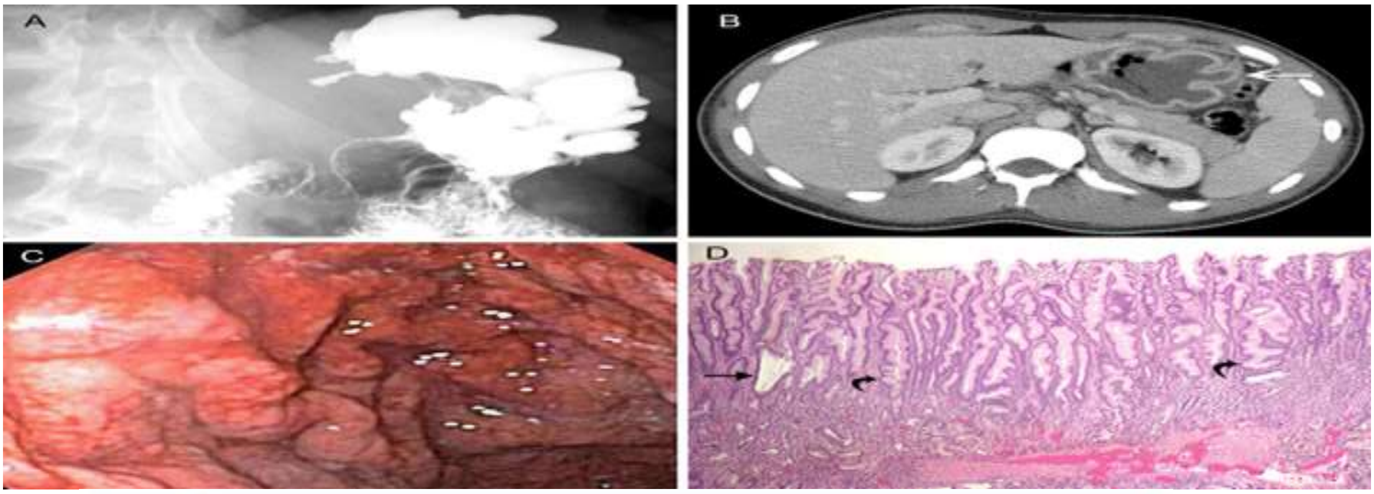
➢ **Chronic gastritis :** classified on the basis of their underlying cause:

1-H.pylori infection: This is the most common cause of chronic gastritis. The predominant inflammatory cells are lymphocytes and plasma cells. It was first discovered in 1982, leading to the identification, description and classification of a multitude of different gastritides.

2-Autoimmune gastritis: This involves the body of the stomach but spares the antrum; it results from autoimmune damage to parietal cells. The histological features are diffuse chronic inflammation, atrophy and loss of fundic glands, intestinal metaplasia and sometimes hyperplasia of enterochromafin-like (ECL) cells. Circulating antibodies to parietal cell and intrinsic factor may be present. In some patients, the degree of gastric atrophy is severe and loss of intrinsic factor secretion leads to pernicious anemia, it is usually associated with other autoimmune disease like thyroid disease. It is precancerous condition.



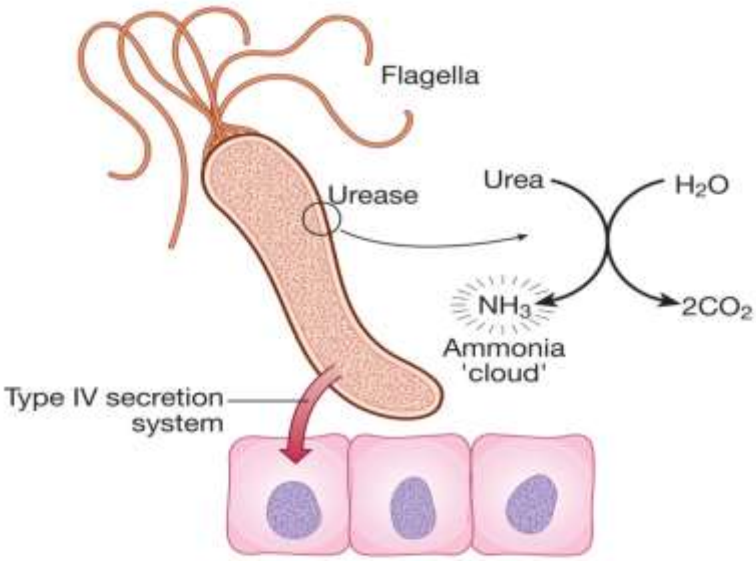
3.Menetrier's disease: In this rare condition, the gastric pits are elongated and tortuous, with replacement of the parietal and chief cells by mucus-secreting cells. The cause is unknown but there is excessive production of transforming growth factor alpha (TGF-α). As a result, the mucosal folds of the body and fundus are greatly enlarged. Most patients are hypochlorhydric. While some patients have upper gastrointestinal symptoms, the majority present in middle or old age with protein-losing enteropathy due to exudation from the gastric mucosa. Endoscopy shows enlarged, nodular and coarse folds, although biopsies may not be deep enough to show all the histological features. Treatment with antisecretory drugs, such as PPIs with or without octreotide, may reduce protein loss and H. pylori eradication may be effective, but unresponsive patients require partial gastrectomy.



i 23.32 Common causes of gastritis
Acute gastritis (often erosive and haemorrhagic)
• Aspirin, NSAIDs • *Helicobacter pylori* (initial infection) • Alcohol • Other drugs, e.g. iron preparations • Severe physiological stress, e.g. burns, multi-organ failure, central nervous system trauma • Bile reflux, e.g. following gastric surgery • Viral infections, e.g. CMV, herpes simplex virus in HIV/AIDS (Ch. 14)
Chronic non-specific gastritis
• *H. pylori* infection • Autoimmune (pernicious anaemia) • Post-gastrectomy
Chronic 'specific' forms (rare)
• Infections, e.g. CMV, tuberculosis • Gastrointestinal diseases, e.g. Crohn's disease • Systemic diseases, e.g. sarcoidosis, graft-versus-host disease • Idiopathic, e.g. granulomatous gastritis
(CMV = cytomegalovirus; HIV/AIDS = human immunodeficiency virus/acquired immunodeficiency syndrome; NSAIDs = non-steroidal anti-inflammatory drugs)

H.pylori infection

- Peptic ulceration is strongly associated with H. pylori infection.
- 50% of people over the age of 50 years are infected in developed countries.
- In developing countries 90% of adults are infected.
- Majority of colonized people remain healthy and asymptomatic.
- Around 90% of duodenal ulcer patients and 70% of gastric ulcer patients are infected with H. pylori
- H. pylori is Gram-negative and spiral, and has multiple flagella at one end, which make it motile, allowing it to burrow and live beneath the mucus layer adherent to the epithelial surface. Here the surface pH is close to neutral and any acidity is buffered by the organism's production of the enzyme urease. It causes chronic gastritis by provoking a local inflammatory response in the underlying epithelium.



The distribution and severity of H. pylori-induced gastritis determine the clinical outcome. In most people, H. pylori causes

1-a mild pangastritis with little effect on acid secretion and the majority develop no significant clinical outcomes.

2- antral-predominant pattern of gastritis (10%) characterised by hypergastrinaemia and a very exaggerated acid production by parietal cells, which could lead to duodenal ulceration.

3-In a much smaller number of infected people, H. pylori causes a corpus-predominant pattern of gastritis **(an early histological marker to identify Helicobacter pylori-infection)** leading to gastric atrophy and hypochlorhydria.